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MINISTRY OF INDUSTRY AND TRADE

COLLEGE OF BUSINESS EDUCATION (CBE)

FIELD PRACTICAL TRAINING REPORT

Submitted in Partial Fulfillment of the Requirement for the Award of Degree in Information Technology

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In embarking on this journey, I would like to begin with a profound sense of gratitude to the Almighty God, whose guidance and blessings have been the cornerstone of my endeavors.

I am also indebted to my lecturers at the College of Business Education for their academic input, and to my fellow students and family for their encouragement and support during the training.

I would like to express my sincere gratitude to the management of Mwananyamala Regional Referral Hospital for granting me the opportunity to undertake my field practical training at their institution. My special thanks go to my field supervisor Mr. Jovan Opanga for the guidance, mentorship, and support provided throughout the attachment period.

SUMMARY/ABSTRACT

This report provides a detailed account of the eight-week field practical training undertaken at Mwananyamala Regional Referral Hospital from 28th July – 19th September 2025 in the ICT Department. The report is divided into several chapters, each offering a unique perspective on this extended field experience.

In the first chapter, there is a detailed overview of the organization structure of Mwananyamala Regional Referral Hospital. This includes a look at how the organization is set up and how it operates.

Chapter two covers a wide range of activities that took place during these 8 weeks. These activities include installing of software (both application and operating systems), carrying out computer preventive maintenance and repair as well as computer troubleshooting to solve errors. Apart from those there were Network based tasks like design and installation of Local Area Networks in other departments offices, termination of Ethernet cables, punching Ethernet cables using RJ-45, network troubleshooting, router configuration (TP Link Routers). Since the organization has different offices with different Information Management System like CMG Portal, BEMIS SIS system, FFARS portal, just to mention a few some activities involved integration with this system like data entry, uploading attachments as well as downloading issued certificates and other legal documents. Thus, this chapter dives into the hands-on experiences gained in a real-world ICT environment as a Computer Science student.

Chapter three is dedicated to the lessons learned the new skilled acquired during the Field Practical Training. Here we explore the practical knowledge and abilities developed, considering, their importance for future endeavors in the computer related fields.

Additionally, another chapter fourth are conclusion and recommendations aimed at improving the university's educational approach. These recommendations are intended to better prepare students for evolving issues in this industry of computer science. The pages at the end of this technical report have included references and appendices.

The report encapsulates an 8-week journey of transformations, providing valuable insights into the real-world applications of Information, Communication and Technology within a public sector organizes. It stands as testament to the personal and professional growth achieved during this extensive field experience.

ABBREVIATIONS

CD -Compact Disc

CMG - Community Microfinance Group Portal

CPU -Central Processing Unit

DVD -Digital Versatile Disc

HDD -Hard Disc Drive

HP - Hewlett Packard

ICT - Information Communication Technology

IT -Information Technology.

ID - Identity Cards

IP -Internet Protocol

LAN -Local Area Network

MIS -Management Information System

CBE - College of Business Education

NIC -Network interface Card

OS -Operating System

PC -Personal Computer

PVC - Polyvinyl Chloride

RAM&ROM -Random Access Memory& Read Only Memory

RJ45 -Register Jack 45

SIS - School Information System

USB -Universal Serial Bus

WAP - Wireless Access Point

WiFi - Wireless Fidelity

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CHAPTER ONE

1.0 Introduction

This is a technical report for Field Practical Training carried at Mwananyamala Regional Referral Hospital in Dar es Salaam from 28th July – 19th September 2025. As a core subject for Degree in Information Technology, it is mandatory to undergo a period of time as practical training on issue learnt, exercise and experience how they relate to lessons learnt at the College.

1.1 Background of Mwananyamala Regional Referral Hospital

Mwananyamala Regional Referral Hospital is located in Kinondoni Municipal Council serving a population of more than 2.2million people, the hospital is under Ministry of Health. It was established in July 1973 as a mother and child Health Clinic (MCH Clinic). In 1983 it was upgraded as the District Hospital, with four wards and bed capacity of 112. Currently hospital bed capacity 254 with bed occupancy of 317. It is a referral hospital for other Kinondoni and Ubungo public and private Health facilities. The hospital attends 1,500 up to 1,800 patients per day.

The hospital plays a critical role in the delivery of healthcare services including surgery, outpatient and inpatient care, pediatrics, obstetrics, laboratory services, and pharmacy. As healthcare delivery increasingly depends on technology, the hospital has adopted ICT systems to support patient data management, hospital administration, and communication. The ICT unit is therefore crucial in ensuring effective and reliable hospital operations.

1.2 Organizational Structure

The hospital operates under the Ministry of Health and is overseen by the Medical Officer In-Charge. It is structured into several departments, including:

- Administration Department
- Medical Services Department
- Nursing Department
- Pharmacy Department
- Laboratory Department
- ICT Unit (responsible for EMR/EHR systems, networking, and technical support)
-Accounting Department

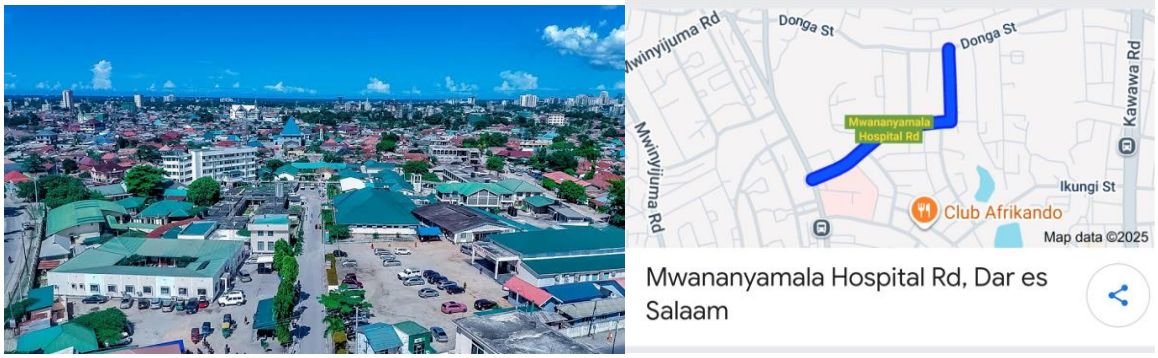


Fig 1.0: A map and image of Mwananyamala RRH

1.3 Objectives of the Field Attachment

The primary objectives of my field attachment were designed to bridge the gap between classroom learning and real-world application within a hospital setting.

1.3.1 To gain practical exposure to ICT operations in a hospital environment.

While theoretical knowledge provides a foundation, actual practice in a healthcare institution enabled me to experience firsthand how information and communication technology supports daily operations such as patient data management, networking systems, and medical record-keeping.

1.3.2 To apply theoretical knowledge acquired during coursework to real-world ICT tasks.

Concepts learned in class, such as database management, networking, system security, and troubleshooting, became more meaningful when applied to actual systems in the hospital. This helped me solidify my understanding and improve my technical competence.

1.3.3 Enhance my problem-solving skills when handling technical challenges.

ICT in healthcare often involves complex systems, and unexpected issues arise frequently. Being on-site gave me the opportunity to learn how to analyze problems, propose solutions, and implement corrective actions in real time. This practical problem-solving experience was invaluable in sharpening my critical thinking abilities.

1.3.4 Develop teamwork and interpersonal communication skills.

Working in a hospital ICT unit required close collaboration with colleagues, healthcare staff, and sometimes even patients. This setting provided a platform to improve my ability to work effectively in a team, listen actively, and communicate technical ideas in a simple and clear manner to non-technical users.

1.3.5 Understand the contribution of ICT to efficient healthcare delivery.

Through my attachment, I observed how ICT systems streamline hospital operations by improving patient record management, enhancing data accuracy, supporting medical decision-making, and facilitating communication across departments. This highlighted the critical role ICT plays in strengthening the quality and efficiency of healthcare services.

CHAPTER TWO

2.0 Field Practical Training Experiences

During the practical training at Mwananyamala RRH different tasks were assigned to be carried as illustrated bellow.

2.1 PC Maintenance.

To make sure that all the systems are working in their good manner in such a way that there will not be any physical fault that might cause the system to fail /collapse /backfire.

Another task that involved maintenance was about cleaning the CPU case by using the blower, repairing the computers keyboard cleaning it to remove the accumulated dust.

Figure 2.0: Showing an opened computer case and a blower machine

2.2 Installation of windows.

The organization had several computers (7 laptops and 16 Desktops) which had to be installed with the newest version windows 10 and some computers that required a change of windows because the window that were running were either corrupt or very outdated and due to security reasons. On top of that performing windows activation on some of the computers there at the facility. In some computers ghost installation was used which including clearing of all data in HDD before installation.

Steps applied in window installation

- ✓ Inserting Bootable Windows 10 pro DVD in DVD ROM Drive or bootable USB.
- ✓ Next select your language, keyboard type (Generally US) and time format
- ✓ Click Install button.
- ✓ Click the box labeled “I agree with the license terms” and click Next to proceed further.
- ✓ In the next screen click on Custom (advanced) option. I hope you are not upgrading here. [If you are doing an upgrade then just clicks on the Upgrade option.]
- ✓ Select the Partition to install a copy was chosen, normally the Operating System installed in drive C.
- ✓ Typed the Computer name, username and password for the user account security.
- ✓ To finalize the installation the option of protecting the computer with recommended options and the GMT zone were chosen. Chose an option of Work Network for using Internet at work place. Therefore, window was successful be installed.

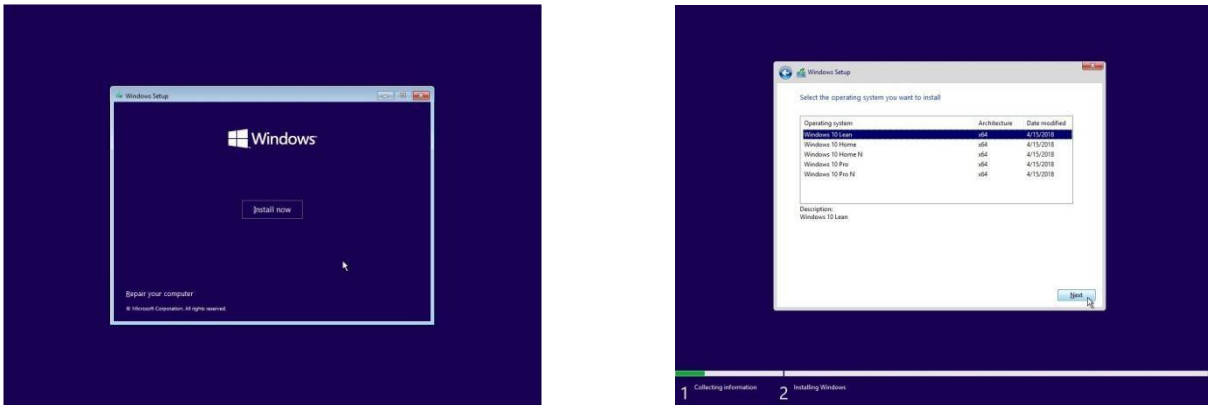


Figure 2.1: Showing some steps during installation of Windows 10

2.3 Installation of Microsoft office

Installation of Microsoft office software and other application in 12 computers after Window installation including Utility programs,

Procedures to follow when we install Microsoft office.

- ❖ Insert Office USB driver into USB port.
- ❖ Once the USB is open, run setup as administrator.

❖ Click Install Now for system with no office or Upgrade for system with low version of Office. Setup will start installation. ❖ Click close to finalize Installation.

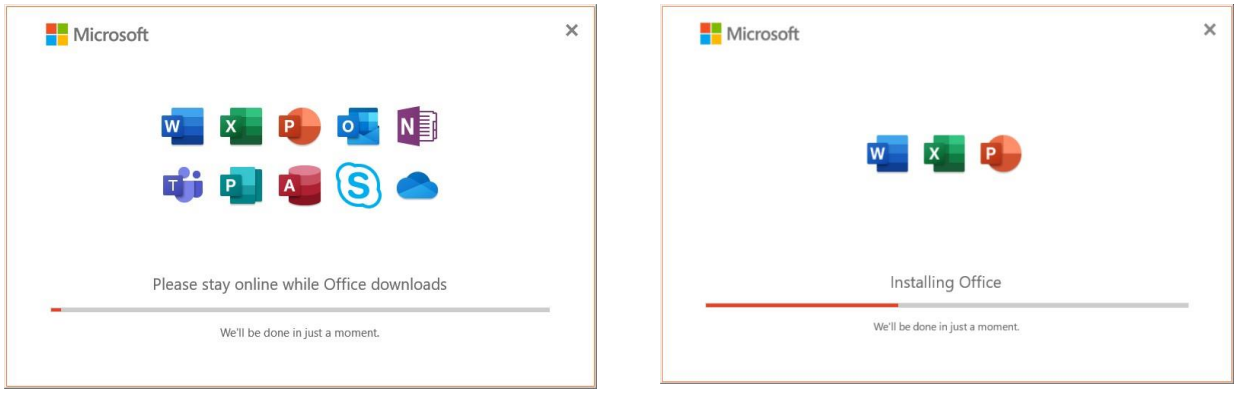


Figure 2.2: Showing installation of Microsoft Office

2.4 Creating table of content.

Through daily departmental reports were required to insert table of content compile every detail prior submission to respective authorities.

How to insert a table of contents: -

Use the built-in heading styles. You can create a table of contents that is based on the custom styles that you will add to your document.

1. **Select the text**, then on the home tab, in the Styles group, **click** will change that text to be a style for the Table of Content
2. Now that your document has styles added to specific entries for your table of contents you are ready to add a Table of Contents.
3. Click where you want to insert the table of contents, at the beginning of a document. If you need to add a blank page at the beginning, click on Insert tab, Blank Page. Have your cursor sitting at the place you want to table to be.

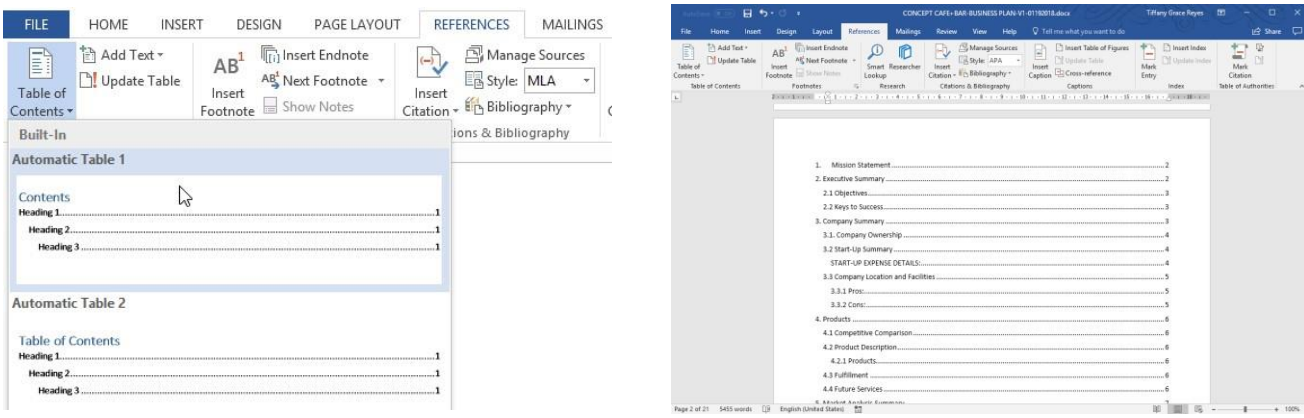


Figure 2.3: Showing how to insert Table of Contents

2.5 Printing a document.

Most of the organization's activities were done through paper work which originated from softcopy to hard copy. Different documents were being printed at our department including Official letters as well as workers Identity Cards (ID).

Steps 1. Click on the File tab and select the Print option to print your current document this will display the Print window options,

2. Along with a preview of the document to the right, such as the range of pages to print.
3. The number of copies to print,
4. After customizing your preferences then **click print to initiate the process**.

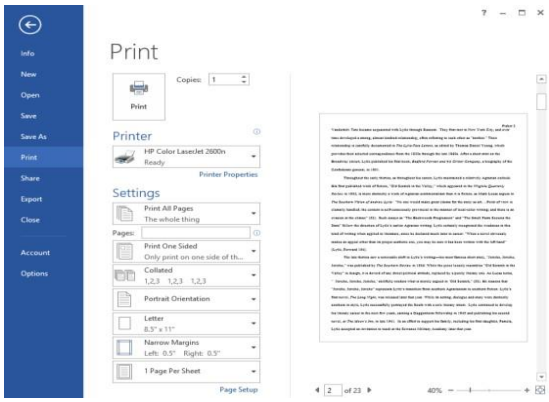


Figure 2.4: Showing Epson printer and a print dialogue box

2.6 Network troubleshooting maintenance and repair

The organization offices rooms have the computers that is wired so as to allow connection around the computer. The Ethernet cables are closed together in a trunk RJ-45 socket the Ethernet cables originating from the faceplates are connected to the computers.

During the practical training among of the activities were troubleshooting to the network that has failed to enable the communications to users.

- The first approach that was used to troubleshoot the connections among the devices is the most basic one that involves **the physical connectivity of the cables to their respective ports**. This is a very basic and common problem that involves physical connectivity of network cable detachment from the network socket (PORT). Sometime the cables fail to link the communications due to other different factors such as wearing and tearing of the cables. For my case I **resolved the network issue by reconfiguring the network cable and replacing them with new ones**. Some of the computer in the system where not able to communicate with server. This was what generally did to solve the issue of

network connectivity for all computers but to reach such a conclusion I had to follow a series of step so that I could easily determine the best solution regarding the network connectivity. For each computer with such problem (Network connectivity Problem) the following were the steps I followed during the Network troubleshooting so as to determine the problem:

1. Ping the gateway IP or the Server IP.
2. If no response, check the network logical configuration such as IP address.
3. Check if the LAN connection is active by checking the LAN LED indicator in the NIC card.
4. Look for any physical damage in the cable, replace the cable if damaged.
5. Check the cable if the cable configuration (straight through or cross-over) is appropriate for that particular connection.
6. Finally Building an Ethernet crossover cable.

The steps were as follows,

- ✓ Determine the cable length required according to the need.
- ✓ Cut off a piece of cable to the desired length using a wire striper normally we remove 5.08cm of the cable jacket.
- ✓ Re-organize the four pairs of twisted cables.
- ✓ Flatten, straighten and line up the cables using the thumb and forefinger.
- ✓ Ensure the cable wire are in the correct order which is white-orange, orange white-green, blue, white-blue, green, white-brown and finally brown.
- ✓ Place the RJ-45 connector on the end of the cable, firmly insert the wires into RJ-45 connector and all the wires should be seen at the end of the connector in their proper positions and finally use the crimping tool to force the wires to hold firmly.

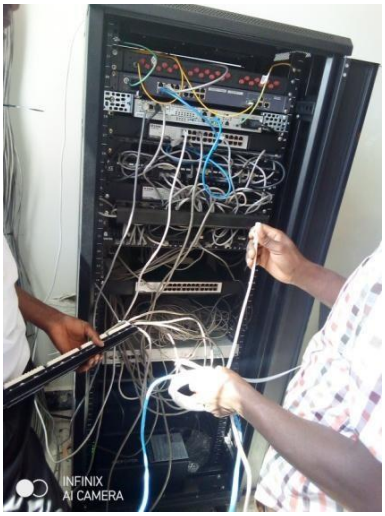


Figure 2.5:Showing network cabinet and a Network Interface Card (NIC)

2.7 Configuration of printer network.

At Mwananyamala RRH there were many printers and most of them are shared amongst with other computer that are connected to the internet because the printers are network printers. In order for them (printers) to communicate they need to be identified by IP address over the network. Each printer is assigned or has its unique IP address that enable it to be easily recognized/identified by computer when a user wants to access printer for printing a particular document.

2.8 Installing of missing system drivers.

Most of the computers in which new operating system was installed faced the problem of missing some important drivers that would make them fully functional. These drivers included network controller and graphic drivers. The Mwananyamala RRH had a copy of universal drivers which were very useful in repairing the missing drivers in the computers and others were directly downloaded from respective official websites.

Figure 2.6: A dialogue box for Driver Updater software

2.9 Antivirus installation, activation and updating.

The type of anti-virus used was Kaspersky antivirus. After the installation of antivirus, another step configured the Anti-Virus for a schedule scan which would take place once a week while the computer is idle and custom scan whenever new mass storage device in connected to the system. and for the case of updating, it was just entering the license product key number to activate online.

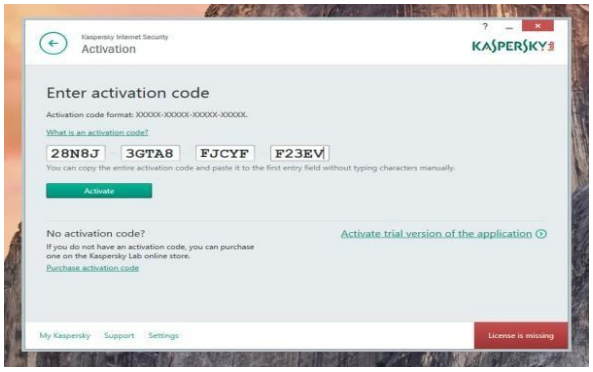


Figure 2.7: An activation window for Kaspersky antivirus software

Steps

During installation the following procedures were followed;

i. The user needs to open the operating system of the laptop and insert CD in the disk drive slot available on the device. ii. Select my computer or my PC folder and open the program with right-click the mouse to start the installation process.

iii. Follow the necessary steps, until the installation will bring the necessary assistance to the user by guiding through the process.

iv. The user should accept/agree the license that show all terms and conditions of the ant-virus program.

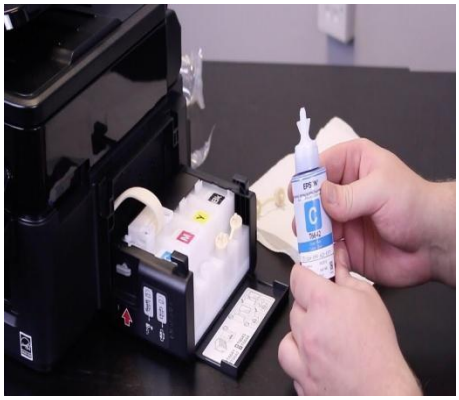
v. The user is required to remove the CD after the installation complete.

vi. The user is required to recommended to restart the laptop so that the new program's setting can properly take place in the operating system of the laptop.

2.10 Troubleshooting Epson L5169 SERIES printers.

Among of the common activities done were troubleshooting of printers like toner cleaning, power cleaning as well as other related settings, ○ Failing to print where **due to lack of enough tonner** where we replaced the worn-out tonner with new tonner.

- The other major issue was **paper jamming** in the printer's paper tray. Open up the printer's hatches so that you can remove the jammed papers from the printer. ○ **Head cleaning,** performed to clear up dogged print head nozzle which causes **faint printing** even if there is enough ink.
- **Nozzle check,** used to check whether the print is working properly by printing a nozzle check print.



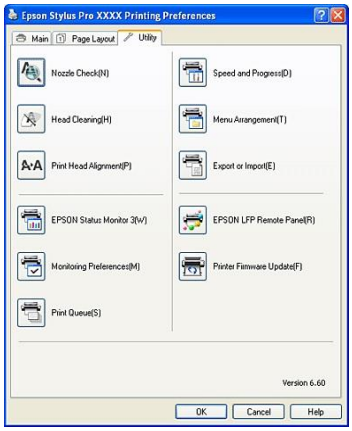


Figure 2.8: Printer ink refilling and printer setting window

2.11 Designing of Mwananyamala RRH workers Identity cards

By using Microsoft publisher and Adobe Photoshop the work of designing the identity cards for workers was done. The followings are tools and equipment used for designing and printing of ID Cards,

- Coloured ink printer (Epson)
- Hard and Soft PVC papers
- Laminator machine
- ID and paper cutter machine (CR80 Size)
- Scanner (scanning signature)
- Microsft Publisher
- Adobe Photoshop

Figure 2.9: Paper cutter, ID cutter machine and Printed ID cards

2.11 Ethernet cable termination

Ethernet cable are also called network cable as are used to connect computers to facilitate communication for various issues like file sharing, resource sharing as well as for internet issues when connected to server.

Straight through is the method that was used during field time because trainee advised that, it is the one mostly used in local area networks. Straight though configuration uses a one wiring standard where as both ends can either use T568A wiring standard or T568B wiring standard was applied the during LAN construction of 4 health facilities for **Afya eHms** frame-work installation.

Figure 2.10: Networking tool box, RJ 45 pin and Router used during network maintenance

CHAPTER THREE

3.1 Key Achievements

During my eight-week field practical training at Mwananyamala Regional Referral Hospital, I was able to accomplish several significant achievements that contributed to both my technical growth and professional development. These experiences not only enhanced my ICT skills but also gave me deeper insight into the role of technology in supporting healthcare services.

3.1.1 Hands-on technical skills.

I actively participated in computer hardware maintenance, which included diagnosing and troubleshooting hardware faults, replacing faulty parts such as RAM modules, hard drives, and power supply units, and configuring printers to ensure smooth operations in various departments. In addition, I strengthened my networking skills through practical tasks such as cable crimping, local area network (LAN) setup, router configuration, and resolving IP conflicts. These activities gave me a strong foundation in handling real-world technical challenges.

3.1.2 Software and systems competence.

I was involved in installing and updating essential applications, including the Hospital Management Information System (HMIS), antivirus programs, and other supporting software. This process not only improved my technical knowledge but also gave me an appreciation of the importance of keeping systems secure and up to date. Furthermore, I gained experience in user account management by creating new profiles for staff, resetting forgotten passwords, and ensuring that only authorized users had access to sensitive systems.

3.1.3 Database and records handling.

I contributed to patient data entry and updating existing records within the HMIS, an experience that emphasized the importance of accuracy and confidentiality in handling medical information. Additionally, I assisted in database backup exercises, which were crucial in ensuring that critical patient records and hospital data were safeguarded against accidental loss or corruption.

3.1.4 Problem-solving and user support

I had the opportunity to assist hospital staff with various ICT-related issues. These included resolving login difficulties, addressing system slowness, and troubleshooting printer connectivity problems. Through these tasks, I developed patience and improved my interpersonal communication skills, especially when dealing with staff who had limited ICT knowledge. This not only enhanced my technical support abilities but also taught me the value of empathy and clear communication in service delivery.

3.1.5 Professional development.

I learned how to collaborate effectively with colleagues in the ICT unit by participating in team-based assignments. Attending departmental meetings further helped me improve my communication and reporting skills, as I was encouraged to share my observations and contribute suggestions. These experiences shaped my confidence and prepared me for future roles that require both technical expertise and strong professional conduct.

3.2 Challenges Encountered

Although my field practical training at Mwananyamala Regional Referral Hospital was highly beneficial and rewarding, I also faced a number of challenges that sometimes-hindered work efficiency and limited my ability to perform certain tasks optimally. These challenges provided important learning experiences and shaped my understanding of the realities of working in a healthcare ICT environment.

3.2.1 Limited ICT resources.

Several departments in the hospital still relied on outdated computers and printers that frequently malfunctioned, which made it difficult to deliver services smoothly. In addition, the hospital often faced shortages of spare parts and modern equipment, which delayed hardware maintenance whenever breakdowns occurred. This limitation highlighted the need for resource planning and investment in ICT infrastructure within the healthcare sector.

3.2.2 Network and power instability.

Occasional power outages and unstable internet connectivity disrupted normal operations, affecting system uptime and delaying critical services. For instance, interruptions sometimes caused delays in patient registration and the updating of medical records in the Hospital Management Information System (HMIS). These challenges emphasized the importance of reliable power supply and stable network connectivity in ensuring efficiency in hospital ICT systems.

3.2.3 Time constraints.

The eight-week period allocated for the field attachment was relatively short compared to the wide range of ICT issues that required attention. While I was able to gain exposure to many areas, certain tasks—such as advanced server management, system security monitoring, and data migration—required more time for in-depth practice and understanding. This limitation underscored the need for extended training opportunities to fully grasp complex ICT operations.

3.2.4 Knowledge gaps

Especially when working with specialized systems like the HMIS. Although I could handle front-end operations such as data entry and user support, I found it difficult to understand the deeper aspects of its back-end database structure and server configurations. For more complex activities like server patching and data migration, I had to rely heavily on guidance from senior ICT staff. This showed me the importance of continuous learning and further training in specialized healthcare ICT systems.

3.2.5 User resistance.

Some hospital staff were reluctant to fully adopt ICT systems, preferring manual record-keeping methods they were more familiar with. This resistance created delays in workflow and increased the workload of the ICT team, who had to spend additional time offering training, guidance, and support. Overcoming this challenge required patience, communication, and persistence to help users appreciate the benefits of digital systems in improving healthcare delivery.

CHAPTER FOUR: Conclusions and Recommendations

4.1 Conclusion

Practical training is a very important program for students to learn and understand the facts about their career. This exposed me to new challenges and opportunities in their profession. I was able to do a lot and experience real working environment which is significant for employment preparation or self-employed upon completion of our studies.

During these periods I understood how to deal with people and different issues concerning my area of professionalism. I also knew how different organizations shape their activities towards the achievement of their goals. Thus, the practical training session is of paramount importance to students as future employers. Moreover, practical training enabled me to understand various fields other than those our career. For instance, at Mwananyamala RRH I learnt many things about administration and information technology, Health and medical issues which are not part of career as an IT but was something in excess and beneficial. I further gained personal confidence and self-trust as I were working with different people having different managerial positions and different qualifications.

During field attachment I observed the following strengths, Mwananyamala RRH a good organizational communication flow which improves the good relationship among the employees. High level of technology that enables Hospital to change with the demand of science technology but the greatest of all I enjoyed working with Mwananyamala RRH and management to meet my field attachment expectations.

4.2 Recommendations

Based on the experience and challenges I encountered during my field practical training at Mwananyamala Regional Referral Hospital, I propose several recommendations aimed at improving and strengthening the field practical training programme for future students. These recommendations focus on extending training opportunities, improving supervision, and ensuring alignment between academic learning and practical exposure.

4.2.1 Extend the training period.

The current eight-week duration is too short for students to gain comprehensive exposure to all aspects of ICT practice. Extending the programme to at least twelve weeks would give students enough time to handle more complex assignments, explore advanced ICT systems, and gain deeper practical experience.

4.2.2 Provide clear training schedules.

A structured roadmap, outlining weekly objectives, expected skills, and specific tasks, would guide students throughout their training. This would not only reduce idle time but also ensure progressive learning and skill development in a systematic way.

4.2.3 Assign mentors or supervisors to each student.

By attaching every trainee to a specific ICT officer, students will benefit from closer supervision, consistent guidance, and stronger professional mentorship. This one-on-one relationship would allow students to receive constructive feedback and professional support, enhancing their growth and confidence.

4.2.4 The programme should introduce rotations across different hospital departments.

Allowing students to rotate between units such as Records, Pharmacy, Laboratory, and Administration would broaden their understanding of how ICT supports diverse hospital functions. This exposure would enable students to appreciate the integral role of ICT in the entire healthcare system rather than viewing it as a stand-alone service.

4.2.5 To include regular student feedback and evaluation sessions.

Weekly meetings between supervisors and trainees would help identify challenges early, address knowledge gaps, and adjust tasks to maximize learning outcomes. This continuous evaluation system would ensure that students remain on track and achieve the intended objectives of the programme.

4.2.6 Strengthen collaboration between hospitals and training institutions.

Closer partnerships between universities or colleges and the hospital would ensure that field training objectives align with classroom teaching. This alignment would make sure students are not only practicing what they have learned theoretically but also gaining industry-relevant skills that meet current job market demands.

4.2.7 I recommend that students be given access to more advanced ICT systems.

Introducing them to modern practices such as server virtualization, cloud computing, and hospital cybersecurity would enrich their training experience and better prepare them for the competitive ICT job market. Exposure to these advanced systems would help bridge the gap between academic learning and professional requirements.

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APPENDICES

- Appendix I: Introduction letter for Field Practical Training